KAIZEN MATHS

Lesson starter

Hypothesis Testing and Inference

Normal Mean Tests

Name _____

S1

A sample of size 25 has mean 203.6 marks. The population standard deviation is known to be 20 marks. Test at the 10% level whether the population mean exam score has increased from 200 marks.

S2

A sample of size 81 has mean 59.2 g. The population standard deviation is known to be 6 g. Test at the 1% level whether the population mean packet mass has changed from 60 g.

S3

Use a critical-value approach to test whether the population mean packet mass has decreased from 120 g. A sample of size 36 has mean 118.76 g, with known $\sigma = 8$ g. Use a 10% significance level.

S4

A sample of size 100 has mean 48.5 minutes. The population standard deviation is known to be 10 minutes. Test at the 1% level whether the population mean delivery time has decreased from 50 minutes.

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Exit ticket

Hypothesis Testing and Inference

Normal Mean Tests

Name _____

EA1

A sample of size 49 has mean 76.76 g. The population standard deviation is known to be 6 g. Test at the 10% level whether the population mean packet mass has changed from 75 g.

EA2

Use a critical-value approach to test whether the population mean packet mass has increased from 75 g. A sample of size 64 has mean 75.7 g, with known $\sigma = 6$ g. Use a 10% significance level.

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Exit ticket

Hypothesis Testing and Inference

Normal Mean Tests

Name _____

EB1

A sample of size 36 has mean 78.5 hours. The population standard deviation is known to be 6 hours. Test at the 1% level whether the population mean battery life has decreased from 80 hours.

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Exit ticket

Hypothesis Testing and Inference

Normal Mean Tests

Name _____

EA1

A sample of size 49 has mean 76.76 g. The population standard deviation is known to be 6 g. Test at the 10% level whether the population mean packet mass has changed from 75 g.

EA2

Use a critical-value approach to test whether the population mean packet mass has increased from 75 g. A sample of size 64 has mean 75.7 g, with known $\sigma = 6$ g. Use a 10% significance level.

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Exit ticket

Hypothesis Testing and Inference

Normal Mean Tests

Name _____

EB1

A sample of size 36 has mean 78.5 hours. The population standard deviation is known to be 6 hours. Test at the 1% level whether the population mean battery life has decreased from 80 hours.

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Teacher answers & working

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S1

A sample of size 25 has mean 203.6 marks. The population standard deviation is known to be 20 marks. Test at the 10% level whether the population mean exam score has increased from 200 marks.

Answer:

do not reject H_0

Working:

Let μ be the population mean exam score.

$H_0: \mu = 200$

$H_1: \mu > 200$

Under H_0, the sampling distribution of the sample mean has mean 200 and standard error $\frac{\sigma}{\sqrt{n}}$.

standard error = $\frac{20}{\sqrt{25}}$ = 4.0000

Standardise the observed sample mean.

$z = \frac{\bar{x} - \mu\boxtimes}{\sigma/\sqrt{n}} = \frac{203.6 - 200}{4.0000} = 0.9000$

Because H_1 is >, use the upper tail.

p-value = 0.1841

Compare the p-value with α = 0.1.

0.1841 > 0.1, so do not reject H_0.

There is insufficient evidence at the 10% level to conclude that the population mean exam score has increased from 200 marks.

Teacher answers & working

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S2

A sample of size 81 has mean 59.2 g. The population standard deviation is known to be 6 g. Test at the 1% level whether the population mean packet mass has changed from 60 g.

Answer:

do not reject H_0

Working:

Let μ be the population mean packet mass.

$H_0: \mu = 60$

$H_1: \mu \neq 60$

Under H_0, the sampling distribution of the sample mean has mean 60 and standard error $\frac{\sigma}{\sqrt{n}}$.

standard error = $\frac{6}{\sqrt{81}}$ = 0.6667

Standardise the observed sample mean.

$z = \frac{\bar{x}-\mu\boxtimes}{\sigma/\sqrt{n}} = \frac{59.2-60}{0.6667} = -1.2000$

Because H_1 is two-sided, double the smaller tail probability.

p-value = 0.2301

Compare the p-value with α = 0.01.

0.2301 > 0.01, so do not reject H_0.

There is insufficient evidence at the 1% level to conclude that the population mean packet mass is different from 60 g.

S3

Use a critical-value approach to test whether the population mean packet mass has decreased from 120 g. A sample of size 36 has mean 118.76 g, with known σ = 8 g. Use a 10% significance level.

Answer:

do not reject H_0

Working:

Let μ be the population mean packet mass.

$H_0: \mu = 120$

$H_1: \mu < 120$

For a left-tailed test at α = 0.1, the critical z-value is -1.2816.

critical region: z ≤ -1.2816

Standardise the sample mean.

$z = \frac{118.76-120}{8/\sqrt{36}} = -0.9316$

Compare the test statistic with the critical region.

-0.9316 is not in the critical region, so do not reject H_0.

There is insufficient evidence at the 10% level to conclude that the population mean packet mass has decreased.

Teacher answers & working

Hypothesis Testing and Inference · Normal Mean Tests

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S4

A sample of size 100 has mean 48.5 minutes. The population standard deviation is known to be 10 minutes. Test at the 1% level whether the population mean delivery time has decreased from 50 minutes.

Answer:

do not reject H_0

Working:

Let μ be the population mean delivery time.

$H_0: \mu = 50$

$H_1: \mu < 50$

Under H_0, the sampling distribution of the sample mean has mean 50 and standard error $\frac{\sigma}{\sqrt{n}}$.

standard error = $\frac{10}{\sqrt{100}}$ = 1.0000

Standardise the observed sample mean.

$z = \frac{\bar{x}-\mu\boxtimes}{\sigma/\sqrt{n}} = \frac{48.5-50}{1.0000} =$ -1.5000

Because H_1 is <, use the lower tail.

p-value = 0.0668

Compare the p-value with α = 0.01.

0.0668 > 0.01, so do not reject H_0.

There is insufficient evidence at the 1% level to conclude that the population mean delivery time has decreased from 50 minutes.

Teacher answers & working

Hypothesis Testing and Inference · Normal Mean Tests

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EA1

A sample of size 49 has mean 76.76 g. The population standard deviation is known to be 6 g. Test at the 10% level whether the population mean packet mass has changed from 75 g.

Answer:

reject H_0

Working:

Let μ be the population mean packet mass.

$H_0: \mu = 75$

$H_1: \mu \neq 75$

Under H_0, the sampling distribution of the sample mean has mean 75 and standard error $\frac{\sigma}{\sqrt{n}}$.

standard error = $\frac{6}{\sqrt{49}}$ = 0.8571

Standardise the observed sample mean.

$z = \frac{\bar{x}-\mu\boxtimes}{\sigma/\sqrt{n}} = \frac{76.76-75}{0.8571} = 2.0500$

Because H_1 is two-sided, double the smaller tail probability.

p-value = 0.0404

Compare the p-value with α = 0.1.

0.0404 ≤ 0.1, so reject H_0.

There is sufficient evidence at the 10% level to conclude that the population mean packet mass is different from 75 g.

EA2

Use a critical-value approach to test whether the population mean packet mass has increased from 75 g. A sample of size 64 has mean 75.7 g, with known σ = 6 g. Use a 10% significance level.

Answer:

do not reject H_0

Working:

Let μ be the population mean packet mass.

$H_0: \mu = 75$

$H_1: \mu > 75$

For a right-tailed test at α = 0.1, the critical z-value is 1.2816.

critical region: z ≥ 1.2816

Standardise the sample mean.

$z = \frac{75.7-75}{6/\sqrt{64}} = 0.9316$

Compare the test statistic with the critical region.

0.9316 is not in the critical region, so do not reject H_0.

There is insufficient evidence at the 10% level to conclude that the population mean packet mass has increased.

Teacher answers & working

Hypothesis Testing and Inference · Normal Mean Tests

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EB1

A sample of size 36 has mean 78.5 hours. The population standard deviation is known to be 6 hours. Test at the 1% level whether the population mean battery life has decreased from 80 hours.

Answer:

do not reject H_0

Working:

Let μ be the population mean battery life.

$H_0: \mu = 80$

$H_1: \mu < 80$

Under H_0, the sampling distribution of the sample mean has mean 80 and standard error $\frac{\sigma}{\sqrt{n}}$.

standard error = $\frac{6}{\sqrt{36}}$ = 1.0000

Standardise the observed sample mean.

$z = \frac{\bar{x}-\mu\boxtimes}{\sigma/\sqrt{n}} = \frac{78.5-80}{1.0000} = -1.5000$

Because H_1 is <, use the lower tail.

p-value = 0.0668

Compare the p-value with α = 0.01.

0.0668 > 0.01, so do not reject H_0.

There is insufficient evidence at the 1% level to conclude that the population mean battery life has decreased from 80 hours.